

Technical Reference: Applied Engineering Optimisation (ENEL445)

Consolidated Study Guide

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1 Mathematical Foundations

1.1 Vector Calculus and Norms

For a column vector $\mathbf{x} \in \mathbb{R}^{n \times 1}$ and a matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, the vector derivative of the objective function $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ with respect to $\mathbf{x}$ is defined as:

$$\frac{\partial f}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^T) \mathbf{x} \quad (1)$$

The **infinity norm** of an $n \times 1$ vector is the maximum absolute value of its entries:

$$\|\mathbf{x}\|_{\infty} = \max_i |x_i| \quad (2)$$

1.2 Taylor Series Expansion

The second-order multivariate Taylor series expansion for a function $f(\mathbf{x})$ around point $\mathbf{c}$ is:

$$f(\mathbf{x}) \approx f(\mathbf{c}) + \nabla f(\mathbf{c})^T (\mathbf{x} - \mathbf{c}) + \frac{1}{2} (\mathbf{x} - \mathbf{c})^T \mathbf{H}(\mathbf{c}) (\mathbf{x} - \mathbf{c}) \quad (3)$$

Where $\mathbf{H}(\mathbf{c})$ is the **Hessian matrix**.

2 Optimality Conditions

2.1 Hessian and Positive Semi-Definiteness

A matrix is **positive semi-definite** if $\mathbf{x}^T \mathbf{A} \mathbf{x} \geq 0$ for all $\mathbf{x}$. In optimisation, the Hessian must be positive semi-definite at a local minimum to ensure the function curves upward in all directions.

2.2 First-Order Necessary Condition (FONC)

For a point $\mathbf{c}$ to be a local minimum of $f(\mathbf{x})$, the directional gradient along an arbitrary direction vector $\mathbf{p}$ must satisfy:

$$\nabla f(\mathbf{c})^T \mathbf{p} = 0 \quad (4)$$

3 Unconstrained Optimisation Logic

3.1 Exact Line Search for Quadratic Functions

Given $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} + \mathbf{b}^T \mathbf{x}$, the optimal step size μ^* along direction $\mathbf{p}$ from point $\mathbf{c}$ is found by solving the unconstrained problem where μ is the design variable. The logic follows:

1. Substitute $\mathbf{x} = \mathbf{c} + \mu\mathbf{p}$ into $f(\mathbf{x})$.
2. Derive the gradient with respect to μ .
3. Solve $\frac{d}{d\mu}f(\mathbf{c} + \mu\mathbf{p}) = 0$ for μ^* .

3.2 Nonlinear Least Squares (Sensor Location)

For an object $\mathbf{u}$ and sensors $\mathbf{s}_i$, the objective is to minimise:

$$\min_{\mathbf{u}} \|\mathbf{r}(\mathbf{u}) - \mathbf{d}\|_2^2 \quad (5)$$

Where $\mathbf{r}(\mathbf{u}) = [r_1(\mathbf{u}), r_2(\mathbf{u}), r_3(\mathbf{u})]^T$ are the true distances $r_i(\mathbf{u}) = \|\mathbf{u} - \mathbf{s}_i\|_2$. The **Jacobian** $\mathbf{J}(\mathbf{u})$ is the matrix of distance gradients.

4 Constrained Optimisation

4.1 Lagrangian and KKT Conditions

For $\min f(\mathbf{x})$ subject to $g(\mathbf{x}) = 0$, the **Lagrangian function** is $\mathcal{L}(\mathbf{x}, \lambda) = f(\mathbf{x}) + \lambda g(\mathbf{x})$. Optimality requires:

- **Stationarity:** $\nabla_{\mathbf{x}}\mathcal{L} = 0$.
- **Primal Feasibility:** $g(\mathbf{x}) = 0$.

4.2 Duality and SQP

- **Lagrangian Dual:** $q(\lambda) = \min_{\mathbf{x}} \mathcal{L}(\mathbf{x}, \lambda)$.
- **Duality Gap:** The difference between primal and dual optima.
- **SQP Logic:** Sequential Quadratic Programming handles the **combinatorial problem** of determining which inequality constraints are **active** using slack variables.

5 Control and Sequential Decision Making

5.1 Dynamic Programming Logic

Recursive Bellman logic for discrete systems $x_{k+1} = u_k$:

```
# Penultimate Step Calculation
J_{h-1}(x_{h-1}) = \min_u [cost(x_{h-1}, u) + J_h(x_h)]
# Reference terminal cost J_h(x_h) = 1
```

5.2 Model Predictive Control (MPC)

For system $x_{k+1} = 0.9x_k + 0.1u_k$ and reference trajectory $\mathbf{r}$:

- **Horizon:** Determined by the length of $\mathbf{r}$ (length 5).
- **Formulation:** $\min \|\mathbf{y} - \mathbf{r}\|_2^2 + \|\mathbf{u}\|_2^2$ subject to dynamics and constraints $|u| \leq 1$.

6 Reinforcement Learning

In Q-Learning, ϵ -greedy exploration manages the trade-off between **exploitation** (current best) and **exploration** (random actions).